

Name of the student: _____ ID. No. _____

Average: 3.73/ 15

Question	Marks	Score
1	6	
2	9	
Total:	15	

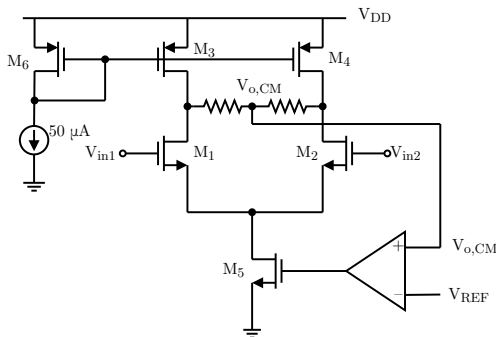
Question 1

Miller compensated two stage differential input single ended output amplifier has DC gain of 80 dB with poles at 15.9 kHz and 740 MHz and zero at 3.18 GHz. Find the gain bandwidth product, phase margin and value of compensation capacitor if load capacitor is 5 pF. [6]

Gain Bandwidth Product in Hz	Phase margin	Compensation capacitor value
159 MHz	75°	1.16 pF

Question 2

For the circuits shown below, the differential gain is 50 V/V without resistive sensing circuit and feedback amplifier. $2V_{ov1}=V_{ov5}$, $3V_{ov1}=|V_{ov3}|$ and $(W/L)_{3,4,6}$ is same. Assume $\lambda_n=0.2\text{ V}^{-1}$ for $L=1\text{ }\mu\text{m}$, $\lambda_p=0.2\text{ V}^{-1}$ for $L=1\text{ }\mu\text{m}$, $\mu_n C_{ox}=150\text{ }\mu\text{A/V}^2$, $\mu_p C_{ox}=50\text{ }\mu\text{A/V}^2$, $V_{th,n}=0.4\text{ V}$, $|V_{th,p}|=0.5\text{ V}$. $L=1\text{ }\mu\text{m}$ for all transistors. and $V_{DD}=1.8\text{ V}$. Neglect body effect. For optimum value of output common mode level $V_{o,CM}$ (to get symmetric output voltage swing), what should be value of V_{REF} ? With optimum $V_{o,CM}$, what is optimum value of $V_{in,CM}$? What should be the common mode gain of amplifier with feedback if $V_{o,CM}$ is allowed to vary by $\pm 1\%$? [9]



Value of V_{REF}	Optimum value of $V_{in,CM}$	Common mode gain with feedback
0.9 V	1 V	0.03 v/v